# Supplementary File 2. Cohort audit

This file supports the main-text section *Cohort assembly and audit* and Table 1 of the main text. Public disease series are frequently supersets, re-analyses or re-runs of one another. Counting such a series as an independent cohort inflates every cross-cohort agreement statistic, because the agreement is partly between a patient and themselves.

## Procedure

For every candidate series we read the sample-level metadata from the GEO series matrix, extracted the subject identifier where the submitters recorded one, and compared identifier sets across series. Where identifiers were absent we compared sample titles, platform, and the exact per-sample value vectors. A series was excluded when its subjects were already present in a series we retained.

## Excluded series (3)

| Accession | Superseded by | Evidence |
|---|---|---|
| GSE30122 | GSE30528, GSE30529 | duplicates GSE30528 + GSE30529 and re-runs 25 of the same controls |
| GSE47183 | GSE104948 | DN subject IDs identical to GSE104948 |
| GSE99340 | GSE104948, GSE104954 | ERCB superset of GSE104948 / GSE104954; ~51 samples are cell lines |

## Residual overlap that could not be removed

64 subjects appear in more than one retained series. These are not duplicate cohorts but the same ERCB patients profiled in two compartments, which is what makes the paired glomerular and tubulointerstitial analyses possible. The benchmark treats the two compartments as separate datasets, so the leave-one-dataset-out folds are not fully independent for those subjects; this is stated in the limitations of the main text.

| Series A | Series B | Shared subjects | Subjects in A | Subjects in B |
|---|---|---|---|---|
| GSE104948 | GSE104954 | 55 | 196 | 195 |
| GSE30528 | GSE30529 | 9 | 22 | 22 |

## Full audit table

| Accession | Omics | Samples loaded | Status | Note |
|---|---|---|---|---|
| GSE30122 | transcriptomics | 69 | excluded | duplicates GSE30528 + GSE30529 and re-runs 25 of the same controls |
| GSE47183 | transcriptomics | 122 | excluded | DN subject IDs identical to GSE104948 |

| Accession | Omics | Samples loaded | Status | Note |
|---|---|---|---|---|
| GSE99340 | transcriptomics | 413 | excluded | ERCB superset of GSE104948/GSE104954; ~51 samples are cell lines |
| GSE1009 | transcriptomics | 6 | retained | |
| GSE104948 | transcriptomics | 196 | retained | |
| GSE104954 | transcriptomics | 195 | retained | |
| GSE111154 | transcriptomics | 8 | retained | |
| GSE131882 | transcriptomics | | retained | |
| GSE142025 | transcriptomics | 36 | retained | |
| GSE142153 | transcriptomics | 40 | retained | |
| GSE175759 | transcriptomics | | retained | |
| GSE20602 | transcriptomics | 18 | retained | |
| GSE30528 | transcriptomics | 22 | retained | |
| GSE30529 | transcriptomics | 22 | retained | |
| GSE96804 | transcriptomics | 61 | retained | |
| ST000691 | metabolomics | | retained | downloaded, not yet analysed |
| ST002145 | metabolomics | | retained | downloaded, not yet analysed |
| ST004442 | metabolomics | | retained | downloaded, not yet analysed |
| ST004483 | metabolomics | | retained | downloaded, not yet analysed |

Samples loaded is the number of samples parsed into the project database. It is zero for assets that were downloaded and inspected but not parsed into the sample table, such as the single-cell objects and most metabolomics studies, which are analysed from their own files.